Wyniki etapu I: Wizja, słownik, model domenowy, reguły biznesowe

<EcoRoute>

Projektowanie oprogramowania

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Wizja i słownik

1. Wprowadzenie

2. Pozycjonowanie

2.1 Sformułowanie problemu

Problem	Lack of awareness and visibility regarding air quality (smog) while moving (cycling, running, walking), leading to unconscious, prolonged exposure to high concentrations of harmful substances.
Dotyczy	Physically active individuals, city dwellers, people with respiratory problems, parents of small children.
Wpływ problemu	Deterioration of health, increased risk of respiratory and circulatory diseases due to excessive exposure to pollution.
Pomyślne rozwiązanie	Reduction of the user's average exposure to pollution, increase in ecological awareness, enabling healthier route planning.

2.2 Opis pozycji produktu

Dla	Active city dwellers (cyclists, runners) and health-conscious individuals
Który/Która	Need real-time air quality information to consciously choose the healthiest route
EcoRoute	Mobile Navigation System (Mobile Application)
Który	Dynamically analyzes the level of pollution on the current route and proposes intelligent, healthier alternatives supported by AI by using sensors located in local area.
Inaczej niż	Standard map applications or static smog maps
Nasz produkt	Offers personalized route recommendations with lower smog levels, taking into account historical data and predictions, not just the current measurement at a single point.

3. Opis udziałowców i użytkowników

3.1 Podsumowanie udziałowców

Nazwa	Opis	Zakres odpowiedzialności
Product Owner	Person responsible for the direction of product development and maximizing value for the customer.	Defines and prioritizes Needs and Features, monitors project progress, approves finished functionalities.

Nazwa	Opis	Zakres odpowiedzialności
Data Provider	External entity supplying air quality data (from monitoring stations and sensors).	Ensures the reliability, timeliness, and correctness of Pollution Measurements.
Development Team	Team responsible for the technical implementation of the application and the AI module.	Ensures system maintenance, performance, scalability, and compliance with quality Requirements.
Marcin Jodłowiec	Mentor and guardian of good academic vibes 😎. Provides guidance, feedback, and ensures project goals are met (and fun is had).	Issuing a grade of 5.0 for the project as the ultimate act of recognition for the team's outstanding craftsmanship. 🏆

3.2 Podsumowanie użytkowników

Nazwa	Opis	Zakres odpowiedzialności
Active User	Individual regularly using navigation for sports or transportation purposes (e.g., cycling, walking).	Uses the application for planning and monitoring routes with health in mind. Provides data about the completed Route (consent to data collection).
Sensitive User	Individual with respiratory illnesses (e.g., asthma) or a parent.	Uses Alternative Route recommendations to minimize exposure to pollution.

4. Opis produktu

4.1 Potrzeby i cechy

Potrzeba	Priorytet	Cechy	Planowane wydanie
To know the current air quality on the path being traveled.	High	Dynamic Pollution Monitoring: Displaying the current pollution level on the map and visual warnings in real time.	1.0
To see what pollution I encountered and how long I was exposed to it.	High	Route Reporting: Generating a summary after route completion with average and maximum pollution levels and exposure time in high-risk zones.	1.0

To choose a healthier route before starting it or next time.	Medium	AI Recommendations for Alternative Routes: Proposing alternative paths with lower forecasted smog levels after route completion.	2.0
To be warned when the pollution level becomes dangerous.	High	Critical Notifications: Push or sound notifications when critical pollution thresholds are exceeded at the current location.	2.0
To save completed routes and their pollution profiles.	Low	Route History: Ability to view and analyze historical route data.	1.0

5. Inne wymagania produktowe

Wymaganie	Priorytet	Planowane wydanie
Platform Requirement: Compatibility with iOS (min. 15.0) and Android (min. 11.0) systems.	High	1.0
Performance Requirement: Response time for a Dynamic Pollution Measurement request along the route must not exceed 2 seconds.	High	1.0
Robustness Requirement: The system must handle a temporary loss of the GPS signal (up to 60 seconds) without interrupting Route recording.	High	1.0
Environmental Requirement: Availability of an external, stable API supplying measurement data for the target operating area.	High	1.0
Legal Requirement: Compliance with GDPR regarding the processing of User geolocation data.	High	1.0

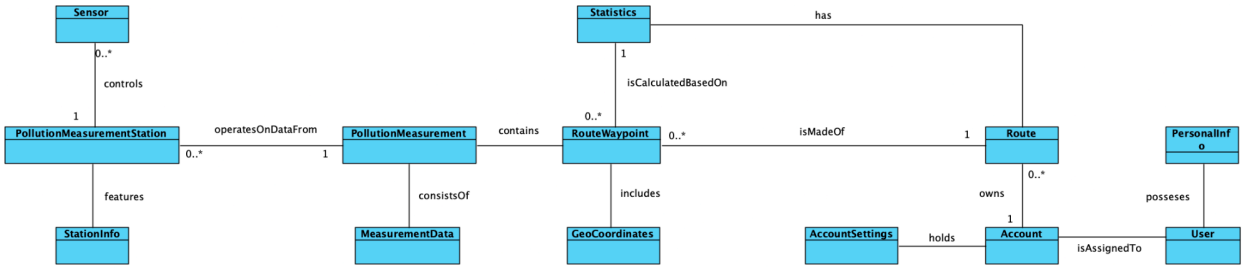
6. Słownik pojęć

Termin	Synonimy	Definicja terminu
User	Runner, Client, Individual Profile	The person using the application; linked to one or more Accounts and associated personal data.
PersonalInfo	Personal Data, Identity Details	A user's identity and regulatory data (e.g., name, contact data, consent flags); possessed by the User.

Account	User Login, Credentials, Application Access	Security and access entity for authentication/authorization; owns Routes and holds AccountSettings.
Account Settings	Preferences, Privacy Settings	Configurable options for an Account (e.g., language, notifications, privacy/consent options).
Route	Track, Path, Activity Session	One recorded activity with start/end time and distance; is made of RouteWaypoints and has Statistics.
Route Waypoint	GPS Point, Location Timestamp, Node	A single point on a Route with a timestamp, includes GeoCoordinates.
GeoCoordinates	Geographic Position, Lattitude/Longtitude	Latitude/longitude pair representing a precise location used by RouteWaypoints.
Statistics	Aggregates, Activity Summary	Computed metrics tied to a Route (e.g., total time, distance, average/peak exposure); calculated from RouteWaypoints.
Pollution Measurement Station	Monitoring Station, Reference Station	Fixed site that collects environmental data; features StationInfo and controls Sensors.
Station Info	Station Metadata, Station Profile	Descriptive and locational metadata about a PollutionMeasurementStation (e.g., address, notes).
Sensor	IoT Device, Measuring Device, Transducer, Detector	Device at a station that measures a specific pollutant; controlled by a PollutionMeasurementStation.
Pollution Measurement	Environmental Reading Set, Air-Quality Dataset	A collection/context of environmental measurements used for analysis; contains RouteWaypoints and consists of MeasurementData.
Measurement Data	Sample, Data Point, Reading	A single measurement record with value, unit and timestamp; belongs to a PollutionMeasurement.
Harmfulness	Toxicity, Adverse Health Effect, Health Risk	The calculated severity of danger posed to the user's body (especially the respiratory and cardiovascular systems) by the measured concentration of pollutants during their activity.

Pollutant Type	Pollutant Category, Measured Substance	An enumerated type (a fixed list) defining the specific hazardous substances that the Sensor is designed to measure (e.g., PM10, NOX, O3).
Start/EndTime	Activity Duration, Time Boundary	The specific datetime values marking the beginning and end of a recorded Route (activity).
Distance	Traveled Length, Covered Ground	The total length of the Route traveled, typically measured in kilometers or miles.
GDPR Consent	Privacy Approval, Data Processing Agreement	A mandatory boolean flag indicating whether the **User** has given permission for their personal data to be stored and processed according to data protection regulations.
Alternative Route	Optimized Path, Low-Pollution Route, Healthier Option	A dynamically calculated or suggested path for a user's activity that minimizes the predicted Exposure to high levels of pollutants compared to a user's standard or a high-traffic route.
Pollution Map	Air Quality Visualization	A dynamic, geographical visualization layer that overlays real-time or forecasted air pollution data, allowing Users to visually assess air quality across different locations.
Exposure Peak	Maximum Exposure Level	The single, highest level of air pollution recorded at any point along the user's Route. It highlights the single most hazardous moment of the activity.
Exposure Average	Mean Exposure Level	The calculated average level of air pollution encountered by the user over the entire duration of the Activity. This metric provides a balanced view of the overall air quality risk for the run.
Health Limit	Safety Threshold, Warning Level	A predefined level of a pollutant's value, established by health organizations (fe. WHO), used to trigger a warning or advisory when the Exposure Peak exceeds it.

Model domenowy



Reguły biznesowe

Reguły biznesowe

1. Structural Rules (Domain Model Constraints)

Id	Rule Description
SR1	Every **Route** must belong to exactly one **Account**; every **Account** must be assigned to exactly one **User**. Hence each **Route** is associated with exactly one **User** via its **Account**.
SR2	Every **MeasurementData** in a **PollutionMeasurement** must have a value greater than or equal to 0.
SR3	The **Status** attribute of a **Route** must take one of the following values: Started or Completed.
SR4	Every **RouteWaypoint** must include exactly one **GeoCoordinates**.
SR5	Every **PollutionMeasurementStation** must feature at least one **Sensor**, every **Sensor** must be attached to exactly one **PollutionMeasurementStation**.

2. Operational Rules (Automation and Calculation)

Id	Rule Description
OR1	If a **Route** is active (status = started) and the **Pollution** value linked to the current **RouteWaypoint** exceeds the configured Health Limit, the system must generate a Critical Warning.
OR2	When a **Route** changes status to Completed, the system must compute and record average and peak pollution statistics for that **Route** based on all associated **PollutionMeasurement** data.
OR3	When a **Route** is marked as Completed, then the AI module must generate at least one **Alternative Route** and populate its projected pollution level attribute (in the related **Statistics**).

3. Decision-Making Rules (Qualification)

Id	Rule Description
DR1	A **Route** may be started (change status to Started) only if GPS is active (current **GeoCoordinates** available) and required permissions are enabled in **AccountSettings**.
DR2	An **Alternative Route** is displayed only if its projected pollution level is at least 10% lower than the completed **Route's** average pollution.

4. Temporal Rules (Time-Based Constraints)

Id	Rule Description
TR1	If a **Route** has status Started and its duration exceeds 8 hours, the system must automatically set status to Completed and finalize **Statistics**.
TR2	**MeasurementData** may be used for dynamic route analysis only if its measurement time is no more than 15 minutes older than the current server time.
TR3	During an active **Route** the system creates a new **RouteWaypoint** (with new **GeoCoordinates**) every 5 seconds.